

Abstract

Deep networks for hyperspectral image (HSI) classification are vulnerable to adversarial attacks. Recent defences such as SACNet^[1] and S³ANet^[2] are effective for standard attacks (FGSM^[3], I-FGSM^[4], PGD^[5], SS-FGSM^[6]), but assume the attacker can directly perturb the pixel values the classifier reads. In real-world scenarios, the physical ground material can be changed before it passes through the atmosphere to the sensors. We propose RTAA, which perturbs ground reflectance and passes it through a differentiable model atmosphere and an imperfection correction step, so the attack accounts for the real sensing pipeline. We evaluate RTAA on two datasets and three classifiers. Under the same perturbation budget, RTAA performs competitively with standard attacks. This provides a physical ground attack for HSI and establishes a realistic threat model for designing sensor-to-classifier defence pipelines.

Introduction

Hyperspectral Imaging (HSI) captures the complete electromagnetic spectrum reflectance at every pixel, consisting of hundreds of narrow colour bands instead of three in RGB. The array of intensity values across these bands is called the spectral fingerprint, which represents the chemical properties of the material, thus allowing distinction between materials that look identical but differ in composition. HSI has applications in food safety, water quality inspections and medical imaging.^[7]

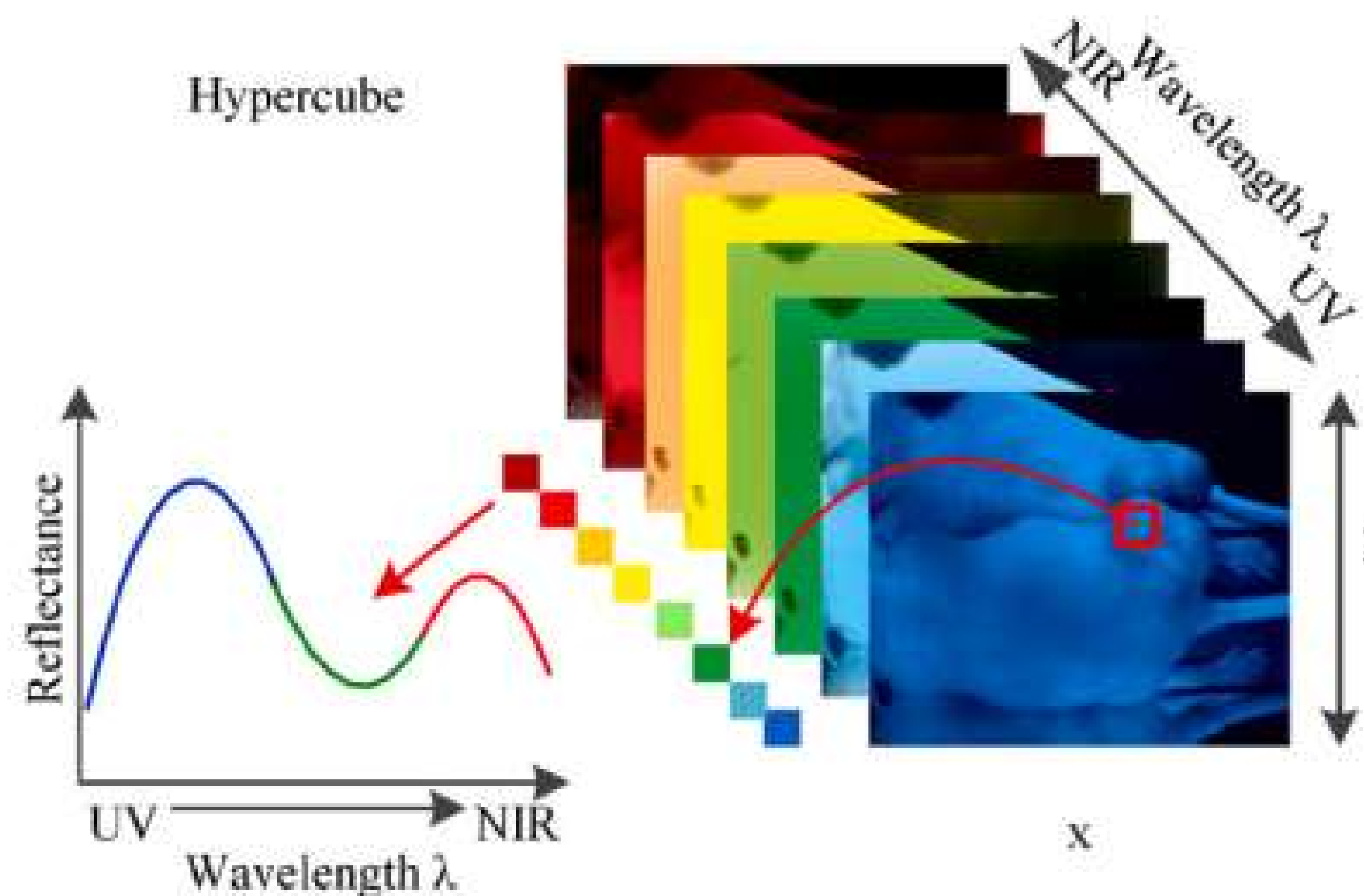


Figure 1. Hyperspectral data cube with two spatial dimensions (x, y) and spectral dimension (λ). Each pixel contains a spectral signature that helps identify materials^[8].

Deep Learning models that classify HSI need to be trustworthy, but they can be fooled by some deliberately crafted, imperceptible changes in the input, called adversarial attack. Existing attacks directly perturb the pixel values read by the classifier. In practice, however, satellites or drones capture reflected sunlight that interacts with the atmosphere and is then corrected before being fed to the classifier. An Attacker can only change the physical scene on the ground instead of changing the pixel values directly. No existing attack evaluates whether classifiers are safe against such realistic physical-world attacks.

Proposed Work

RTAA perturbs the ground reflectance instead of the input values to the classifier. Let the true ground reflectance be R and the perturbations be δ , with ($\|\delta\|_\infty \leq \epsilon$). We model the sensing process as:

$$L_{\text{sensor}}(\lambda) = T_{\text{atm}}(\lambda) R(\lambda) E_{\text{sun}}(\lambda) + L_{\text{path}}(\lambda) \quad (1)$$

The correction pipeline uses an estimated atmosphere, which is never exactly right:

$$\text{atm}_{\text{assumed}} = \text{atm}_{\text{true}} + \text{bias} + \mathcal{N}(0, \sigma^2) \quad (2)$$

Using the wrong atmosphere to invert Eq. (1) gives a recovered reflectance $\hat{R}$, which is not identical to $R + \delta$. The recovered reflectance is projected through a differentiable PCA and fed to the classifier. The complete pipeline is differential, so gradients can be computed w.r.t δ , which is updated using a momentum-based, multi-step gradient attack. We also constrain δ using a physical-viability loss to ensure the perturbed reflectance remains physically realisable.

Highlights of our Proposed Work

- Targets the real-world scene instead of the classifier's input, testing a threat model not previously evaluated.
- If the assumed and true atmosphere were identical, the whole forward and correction loop would cancel out, reducing to a standard pixel attack.
- Under the same perturbation budget, RTAA matches PGD^[5] on SACNet and S3ANet across two datasets, even with added physical constraints.

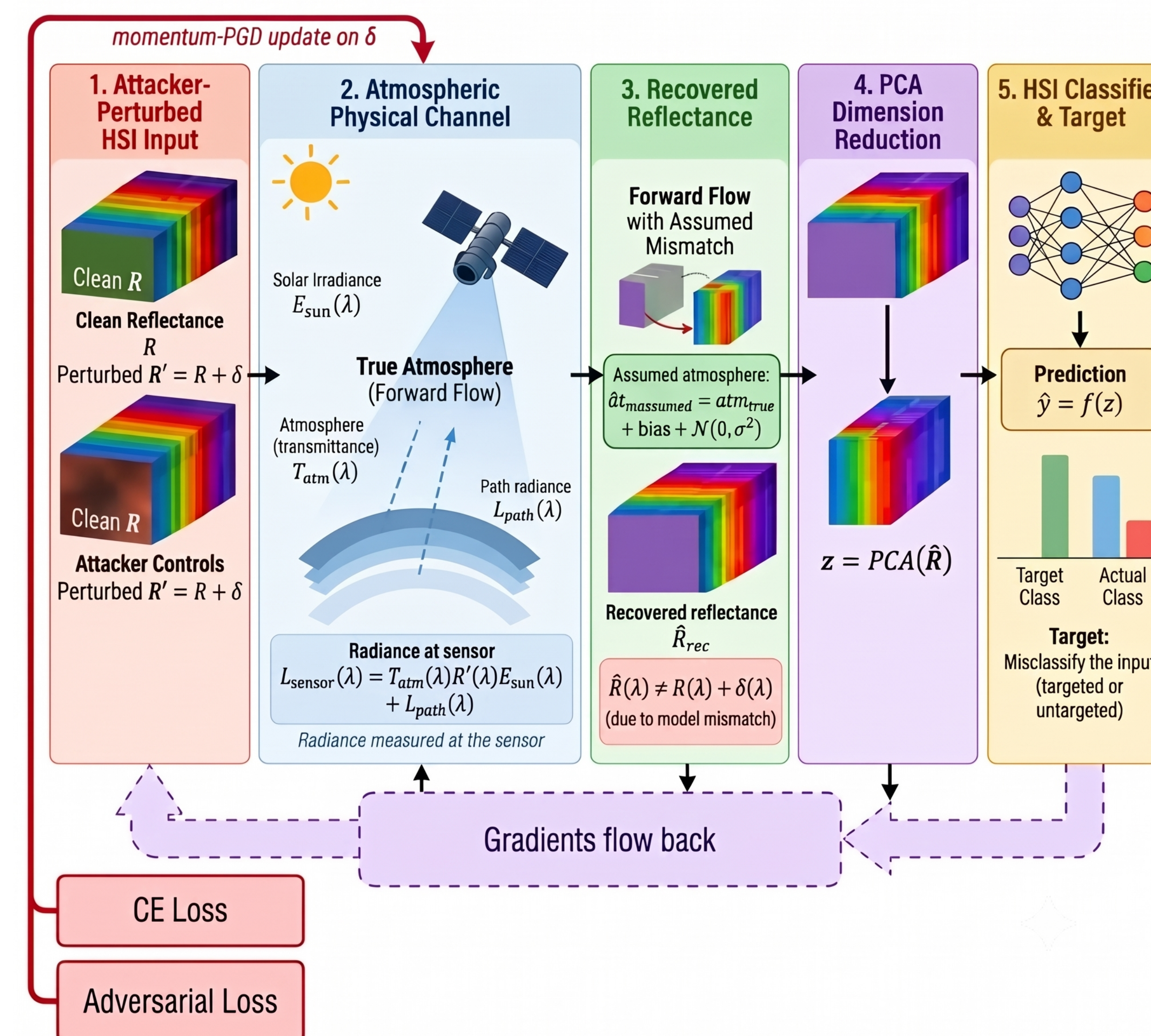


Figure 2. RTAA pipeline. A perturbation δ is introduced into the ground reflectance R , after which the perturbed signal passes through the atmospheric forward model and an imperfect atmospheric correction. The recovered reflectance is then transformed using PCA and classified by the HSI model. The resulting classification loss is propagated backward through the differentiable pipeline to update δ using momentum-PGD.

Results

Table 1. Attack success rate (ASR %) of FGSM^[3], I-FGSM^[4], PGD^[5], SS-FGSM^[6] against two defences (S3ANet^[2], SACNet^[1]) and a transformer classifier (SpectralFormer^[9]) compared to our proposed RTAA. All attacks are performed at the same budget $\epsilon = 0.05$.

Dataset	Model	FGSM	I-FGSM	PGD	SS-FGSM	RTAA
Pavia University	S3ANet	52.20	70.51	88.85	63.05	88.81
	SACNet	82.25	87.32	89.86	80.80	89.49
	SpectralFormer	75.00	85.33	89.27	86.33	92.33
Salinas	S3ANet	35.79	38.13	62.54	60.55	63.54
	SACNet	76.84	59.30	68.77	75.79	78.95
	SpectralFormer	77.59	92.33	96.67	94.33	97.08

Table 2. ASR (%) of PGD and RTAA under different perturbation budgets (ϵ) on defense models.

Dataset	Model	Attack	$\epsilon=0.01$	$\epsilon=0.02$	$\epsilon=0.05$	$\epsilon=0.10$	$\epsilon=0.20$
Pavia University	S3ANet	PGD	13.90	46.49	88.85	97.64	100.00
		RTAA	14.20	49.10	88.81	98.31	100.00
	SACNet	PGD	50.36	73.36	89.86	94.93	95.93
		RTAA	51.81	75.55	89.49	94.57	97.10
Salinas	S3ANet	PGD	7.02	25.75	62.88	94.65	99.33
		RTAA	6.35	23.08	63.54	91.97	99.27
	SACNet	PGD	43.51	51.58	68.77	88.77	89.12
		RTAA	44.56	64.21	78.95	88.07	87.37

Conclusion and Future Work

RTAA shows that defences built to withstand direct digital noise are not necessarily safe for real-world deployment, as attackers can exploit the physical scene. By modelling atmospheric effects and sensor corrections, our approach provides a more realistic evaluation of HSI classifiers.

In future work, we plan to validate RTAA under varying atmospheric conditions using simulated atmospheric data from MODTRAN^[10], evaluating patch-based attacks, extending to more datasets (e.g. Indian Pines, Houston) and testing on newer backbone models such as Mamba-based and lightweight classifiers.

References

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